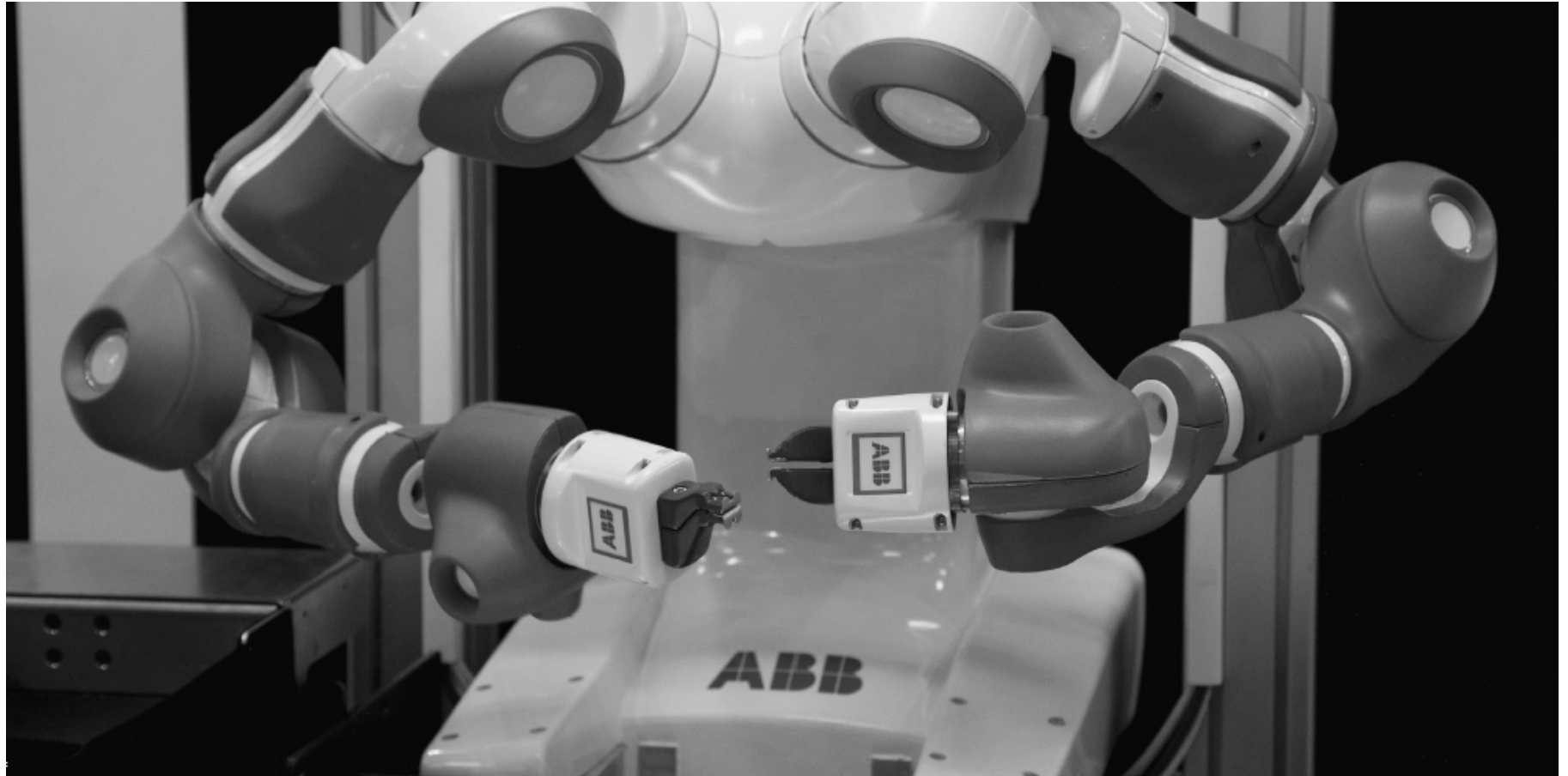




2

Accounting and Engineering Economy
Adapted from Newman et., Oxford Press

The Two Faces of ABB



The Two Faces of ABB, cont'd

- ABB Ltd was a world leader in the power and automation technology areas in the late 1990s
- Assigned overhead costs to its divisions on equal basis although some activities incurred more costs than others
- Switched to activity-based costing (ABC), which assigns costs to the activities that actually incur them
- One of its serious problems arose at Combustion Engineering (an American subsidiary) due to numerous asbestos liability claims
- ABB also issued poor third-quarter earnings report after assuring about being on target to improve earnings and decrease debt
- The price of ABB shares nose-dived and credit rating agencies viewed the company's bonds poorly
- ABB recovered, and as of December 2016 was one of the largest engineering conglomerates in the world

Learning Objectives

- Describe the links between engineering economy and accounting
- Describe the objectives of general accounting, explain what financial transactions are, and show how they are important
- Use a firm's balance sheet and associated financial ratios to evaluate the firm's health
- Use a firm's income statement and associated financial ratios to evaluate the firm's performance
- Use traditional absorption costing to calculate product costs
- Understand the greater accuracy in product costs available with ABC

The Role of Accounting

- Engineering economy focuses on the financial aspects of projects, whereas accounting focuses on the financial aspects of firms
- Three functions within businesses:
 - 1) **Engineering economics:** Analyzes economic impact of alternatives and projects over their life cycles
 - 2) **Accounting:** Determines the dollar impact of past decisions, reports on the economic viability of a unit or firm, and evaluates potential funding sources
 - 3) **Management:** Allocates available investment funds to projects, evaluates unit and firm performance, allocates resources, and selects and directs personnel

The Role of Accounting, cont'd

- Accounting for Business Transactions
 - A business transaction involves two parties and the exchange of dollars (or the promise of dollars) for a product or service
 - Accounting functions:
 - Records, analyzes, and reports on transactions
 - Summarizes and analyzes financial data
 - Provides data for general accounting and cost accounting

The Balance Sheet

- Primary accounting statements:
 - Balance sheet:
 - A firm's financial condition at a specific time
 - Lists a firm's assets, liabilities, and equity
 - Income statement:
 - A firm's performance over a period of time

The Balance Sheet, cont'd

Balance Sheet for Engineered Industries, December 31, 2016 (all amounts in \$1,000s)

Assets		Liabilities	
Current assets		Current liabilities	
Cash	1,940	Accounts payable	1,150
Accounts receivable	950	Notes payable	80
Securities	4,100		
Inventories	1,860	Accrued expense	950
(minus)Bad debt provision	<u>-80</u>	Total current liabilities	<u>2,180</u>
Total current assets	8,770		
		Long-term liabilities	<u>1,200</u>
		Total liabilities	3,380
Fixed assets			
Land	335		
Plant and equipment	6,500		
(minus)Accumulated depr.	<u>-2,350</u>		
Total fixed assets	4,485		
Other assets		Equity	
Prepays/deferred charges	140	Preferred shares	110
Intangibles	<u>420</u>	Common shares	650
Total other assets	560	Capital surplus	930
		Retained earnings	<u>8,745</u>
		Total equity	<u>10,435</u>
Total assets	13,815	Total liabilities and equity	13,815

FIGURE 2-1 Sample balance sheet.

The Balance Sheet, cont'd

- **Assets:** Owned by the firm and have monetary value
- **Liabilities:** Dollar claims against the firm
 - Short-term (current): Due within one year
 - Long-term: Longer due dates (mortgages, bonds, loans, etc.)
- **Equity (or net worth):** represents funding from the firm and its owners (the shareholders)
 - At any point in time the difference between the total worth minus what is owed
- Fundamental accounting equation:
 - $\text{Assets} = \text{Liabilities} + \text{Equity}$

The Balance Sheet, cont'd

- For Engineered Industries
 - Total current assets are \$13,815,000
 - Total current and long-term liabilities are \$2,180,000 and \$1,200,000, respectively
 - Total equity is \$10,435,000
- Fundamental accounting equation:
 - $\text{Assets} = \text{Liabilities} + \text{Equity}$
 - $\$13,815,000 = \$3,380,000 + \$10,435,000$

The Balance Sheet, cont'd

- Working Capital
 - Working capital = Current assets – Current liabilities
 - For Engineered Industries, there would be
 $\$8,770,000 - \$2,180,000 = \$6,590,000$ available in working capital

The Balance Sheet, cont'd

- Financial Ratios Derived from Balance Sheet Data
 - Firm's health can be evaluated through ratios of quantities on the balance sheet
 - Current ratio = Current assets/Current liabilities
 - Shows the ability of a firm to cover current liabilities
 - Historically, firms have aimed to be at or above a ratio of 2.0
 - Acid-test ratio (or quick ratio) = Quick assets/Total current liabilities
 - Shows the ability of a firm to pay debt quickly
 - Current inventories are excluded from quick assets due to the time required to sell them

The Balance Sheet, cont'd

- Financial Ratios Derived from Balance Sheet Data, cont'd
 - For Engineered Industries
 - The current ratio is above 2 ($8,770,000 / 2,180,000 = 4.02$)
 - The acid-test ratio is well above 1 [$(1,940,000 + 950,000 + 4,100,000) / 2,180,000 = 3.21$]

The Income Statement

- Also known as the profit and loss statement
- Summarizes the firm's revenues and expenses over a period of time (month, quarter, year, etc.)
- Used to evaluate revenue and expenses that occur in the interval between consecutive balance sheet statements

The Income Statement, cont'd

Income Statement for Engineered Industries for End of Year 2016 (all amounts in \$1,000)

Operating revenues and expenses**Operating revenues**

Sales	18,900
(<i>minus</i>) Returns and allowances	<u>-870</u>
Total operating revenues	18,030

Operating expenses**Cost of goods and services sold**

Labour	6,140
Materials	4,640
Indirect cost	2,280
Selling and promotion	930
Depreciation	450
General and administrative	2,160
Lease payments	<u>510</u>
Total operating expense	17,110

Total operating income	920
-------------------------------	------------

Non-operating revenues and expenses

Rents	20
Interest receipts	300
Interest payments	<u>-120</u>

Total non-operating income	200
-----------------------------------	------------

Net income before taxes	1,120
--------------------------------	--------------

Income taxes	<u>390</u>
--------------	------------

Net profit (loss) for Year 2016	730
--	------------

FIGURE 2-2 Sample income statement.

The Income Statement, cont'd

- The income statement reports the firm's net income (profit) or loss by subtracting expenses from revenues
 - $\text{Revenues} - \text{Expenses} = \text{Net profit (Loss)}$
- If revenues minus expenses is positive, there has been a profit; if it is negative, a loss has occurred

The Income Statement, cont'd

- For Engineered Industries
 - Total revenues = \$18,350,000
 - Total expenses = \$17,230,000
 - Net income = Revenues - Expenses
 - = \$18,350,000 - \$17,230,000
 - = \$1,120,000 before taxes (\$730,000 after taxes)

The Income Statement, cont'd

- For Engineered Industries
 - Total revenues = \$18,350,000
 - Total expenses = \$17,230,000
 - Net income = Revenues - Expenses
 - = \$18,350,000 - \$17,230,000
 - = \$1,120,000 before taxes (\$730,000 after taxes)

The Income Statement, cont'd

- Financial Ratios Derived from Balance Sheet Data
 - Net Profit Ratio = Net profit/Net sales revenue
 - Indicates cost efficiency of operations and firm's ability to convert sales into profits
 - Net sales revenue equals sales minus returns and allowances
 - Interest Coverage Ratio = Total Income/Interest Payments
 - Indicates how much revenue must drop to affect the firm's ability to finance its debt
 - It should be at least 3.0 for industrial firms
 - Total income is total revenues minus all expenses except interest payments

The Income Statement, cont'd

- Financial Ratios Derived from Balance Sheet Data, cont'd
 - Net Profit Ratio = Net profit/Net sales revenue
 - For Engineered Industries, the net profit ratio is $730,000/18,030,000 = 0.040 = 4.0\%$
 - Interest Coverage Ratio = Total Income/Interest Payments
 - Engineered Industries has an interest coverage ratio of $(18,350,000 - 17,110,000)/120,000 = 10.3$

The Income Statement, cont'd

- Linking the Balance Sheet, Income Statement, and Capital Transactions
 - There are many links between the balance sheet, income statement, and cash flows that make up business transactions
 - For engineering economic analysis, the following links are the most important:
 - Overall profit or loss (income statement) and the starting and ending equity (balance sheets)
 - Acquisition of capital assets
 - Depreciation of capital assets

The Income Statement, cont'd

- Linking the Balance Sheet, Income Statement, and Capital Transactions, cont'd
 - The overall profit or loss during the year (shown on the income statement) is reflected in the change in retained earnings (RE) between the balance sheets at the beginning and end of the year
 - To find the change in retained earnings (RE), one must also subtract any dividends distributed to the owners and add the value of any new capital stock sold
 - Change in Retained Earnings (RE):
 - $RE_{\text{end}} = RE_{\text{beg}} + \text{Net Income/Loss} + \text{New Stock} - \text{Dividends}$

The Income Statement, cont'd

EXAMPLE 2-1

For simplicity, assume that Engineered Industries will not pay dividends in 2015 and did not sell any capital equipment. It did purchase \$4,000,000 in capital equipment. Use the linkages just described to decide what can be said about the values on the balance sheet at the end of 2016.

SOLUTION

First, the net profit of \$730,000 will be added to the retained earnings from the end of 2015 to find the new retained earnings at the end of 2016:

$$RE_{12/31/2012} = \$730,000 + \$8,745,000 = \$9,475,000$$

Second, the fixed assets shown at the end of 2015 would increase from \$6,500,000 to \$6,900,000. (*Note:* That is a major investment of the retained earnings in the firm's physical assets.)

Third, the accumulated depreciation would increase by the \$450,000 in depreciation shown in the 2016 income statement from the \$2,350,000 shown in the 2015 balance sheet. The new accumulated depreciation on the 2016 balance sheet would be \$2,800,000. Combined with the change in the amount of capital equipment, the new fixed asset total for 2016 would equal

$$\$335,000 + \$6,900,000 - \$2,800,000 = \$4,435,000$$

Traditional Cost Accounting

- **Cost accounting:** Used to develop product costs, determine the mix of labour, materials, and other costs in a production setting, and to evaluate outsourcing and subcontracting possibilities

Traditional Cost Accounting, cont'd

- Direct and Indirect Costs
 - Costs incurred to produce a product or service are traditionally classified as either direct or indirect (overhead)
 - **Direct costs:** Activities directly associated with the final product or service produced (e.g., materials, labour, etc.)
 - **Indirect costs:** Costs not easily linked directly to individual products or services (e.g., machine depreciation, management/sales/administration costs, etc.)

Traditional Cost Accounting, cont'd

- Indirect Cost Allocation
 - Indirect costs are sometimes allocated across projects through “absorption costing”
 - Allocating overhead costs based on factors such as number of direct-labour costs each project has or direct-labour hours or direct-material cost or total direct cost

Traditional Cost Accounting, cont'd

- Problems with Traditional Cost Accounting
 - Indirect and overhead costing can distort product costs and the decisions based on those costs
 - Some firms are shifting to activity-based costing (ABC), where activities are linked to specific costs and in this way overhead is potentially minimized

Traditional Cost Accounting, cont'd

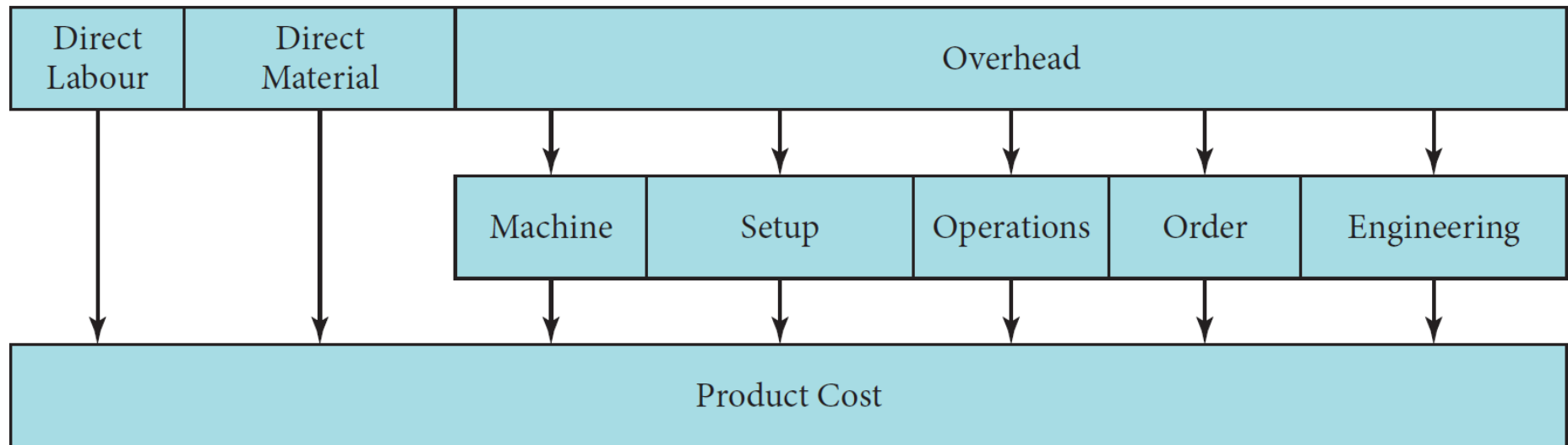


FIGURE 2-3 Activity-based costing versus traditional overhead allocation.

Other Problems to Watch For

- Centralized accounting systems have often been accused by project managers of being too slow or “untimely”
- If an organization establishes numerous files and systems so that stakeholders have the timely data they need, the level of accuracy in one or all systems may be low
 - Analysts making cost estimates will have to consider other internal data sources.
- Inventory or land valued too low because its based on acquisition cost
- Capital equipment being valued too high or too low depending on the depreciation methods and company policy